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22.

On the Intersections, Contacts, and Other Correlations of Two Conics Expressed by Indeterminate Coordinates

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p. 119

Let $U = 0$, $V = 0$ be two homogeneous equations of the second degree with real coefficients, between the same three variables ξ, η, ζ.

The direct and most general mode of determining the intersections of the conics expressed by these equations would be to make

$$a\xi + b\eta + c\zeta = t, \qquad a'\xi + b'\eta + c'\zeta = u:$$

eliminating ξ, η, ζ between the four equations in which they appear, there results a biquadratic equation between t and u. The nature of the intersections will depend upon the nature of the roots of this biquadratic; and thus the conditions may be expressed analytically, which will represent the several cases of all the intersections being real or all imaginary, or one pair real and the other imaginary. These analytical conditions will depend upon the signs of certain functions of the coefficients of the given and the assumed equations being of an assigned character; my endeavour has been to obtain conditions of a character perfectly symmetrical and free from the coefficients arbitrarily introduced.

In this research I have only partially succeeded, but the method employed, and some of the collateral results, will, I think, be found of sufficient interest to justify their appearance in the pages of this *Journal.*

Adopting Mr Cayley's excellent designation, let the four points of intersection of the two conics be called a quadrangle. This quadrangle will have three pairs of sides; the intersections of each pair, from principles of analogy, I call the vertices of the quadrangle. Then, inasmuch as the four p. 120

sets of ratios $\xi : \eta : \zeta$, corresponding with the four sets of the ratio $t : u$, must be so related that we may always make

$$\begin{aligned}
\frac{\xi_1}{\zeta_1} &= a + b\sqrt{-1}, & \frac{\eta_1}{\zeta_1} &= c + d\sqrt{-1}, \\
\frac{\xi_2}{\zeta_2} &= a - b\sqrt{-1}, & \frac{\eta_2}{\zeta_2} &= c - d\sqrt{-1}, \\
\frac{\xi_3}{\zeta_3} &= \alpha + \beta\sqrt{-1}, & \frac{\eta_3}{\zeta_3} &= \gamma + \delta\sqrt{-1}, \\
\frac{\xi_4}{\zeta_4} &= \alpha - \beta\sqrt{-1}, & \frac{\eta_4}{\zeta_4} &= \gamma - \delta\sqrt{-1},
\end{aligned}$$

we may easily draw the following conclusions.

If all the four points of the quadrangle of intersection are real, the three vertices and the three pairs of sides are all real. If only two points of the quadrangle are real, one vertex and one of the three pairs of sides will be real; the other two vertices and two pairs of sides being imaginary. If all four points of the quadrangle are unreal, one pair of sides will be real and the other two pairs imaginary, as in the last case; but all the three vertices will remain real, as in the first case. Hence we have a direct and simple criterion for distinguishing the case of *mixed* intersection from intersection wholly real or wholly imaginary; namely, that the cubic equation of the roots of which the coordinates of the vertices are real linear functions shall have *a pair of imaginary roots*. This is the sole and unequivocal condition required.

The equation in question is, or ought to be, well known to be the determinant in respect to ξ, η, ζ of $\lambda U + \mu V$. In fact, if we write

$$\begin{aligned}
U &= a\xi^2 + b\eta^2 + c\zeta^2 + 2a'\eta\zeta + 2b'\zeta\xi + 2c'\xi\eta,\\
V &= \alpha\xi^2 + \beta\eta^2 + \gamma\zeta^2 + 2\alpha'\eta\zeta + 2\beta'\zeta\xi + 2\gamma'\xi\eta,\\
\lambda U + \mu V &= (a\lambda + \alpha\mu)\xi^2 + \&c.\\
&= A\xi^2 + B\eta^2 + C\zeta^2 + 2A'\eta\zeta + 2B'\zeta\xi + 2C'\xi\eta,
\end{aligned}$$

the ratios of the coordinates ξ, η, ζ of the vertex of $\lambda U + \mu V$ may easily be shown to be identical with

$$AB - C'^2 : C'A' - B'B : B'C' - A'A,$$

and will be real or imaginary as $\lambda : \mu$ is one or the other.

If then the cubic equation in $\lambda : \mu$, namely, $\square_{\xi\eta\zeta}(\lambda U + \mu V) = 0$, has a pair of imaginary roots, that is, if

$$\square_{\lambda\mu}\square_{\xi\eta\zeta}(\lambda U + \mu V)$$

is a positive quantity, the intersections of U and V are of a mixed kind, that is, the two conics have two real points in common. p. 121

I may remark here, *en passant*, that if we form the biquadratic equation in t and u, $\phi(t, u) = 0$ from the equations

$$U = 0, \qquad V = 0, \qquad a\xi + b\eta + c\zeta = t, \qquad a'\xi + b'\eta + c'\zeta = u,$$

and if any reducing cubic of this equation be $P(\theta, \omega) = 0$, the determinant of $P(\theta, \omega)$ must, from what has been shown above, be identical with

$$\square_{\lambda\mu}\square_{\xi\eta\zeta}(\lambda U + \mu V)$$

multiplied by some squared function of the extraneous coefficients

$$a, b, c; \quad a', b', c'.$$

If $\square\square(\lambda U + \mu V)$ is a negative quantity, it remains to distinguish between the cases of the conics intersecting really in four points or not at all.

The most obvious mode of proceeding to distinguish between purely real and purely imaginary intersections would be as follows. Let $\lambda_1, \mu_1; \lambda_2, \mu_2; \lambda_3, \mu_3$ be the three sets of values of λ, μ which satisfy the equation

$$\square(\lambda U + \mu V) = 0$$

and make

$$\begin{array}{lll} A_1 = a\lambda_1 + \alpha\mu_1, & A_2 = a\lambda_2 + \alpha\mu_2, & A_3 = a\lambda_3 + \alpha\mu_3, \\ C_1 = c\lambda_1 + \gamma\mu_1, & C_2 = c\lambda_2 + \gamma\mu_2, & C_3 = c\lambda_3 + \gamma\mu_3, \\ B'_1 = b'\lambda_1 + \beta'\mu_1, & B'_2 = b'\lambda_2 + \beta'\mu_2, & B'_3 = b'\lambda_3 + \beta'\mu_3, \\ A_1C_1 - B'^2_1 = e_1, & A_2C_2 - B'^2_2 = e_2, & A_3C_3 - B'^2_3 = e_3. \end{array}$$

Now if the equation

$$A\xi^2 + B\eta^2 + C\zeta^2 + 2A'\eta\zeta + 2B'\zeta\xi + 2C'\xi\eta = 0$$

represent a pair of straight lines, it may be thrown into the form

$$Au^2 + \frac{AC - B'^2}{A} v^2 = 0,$$

where u and v are linear functions of ξ, η, ζ, and the straight lines will be real or imaginary, according as $B'^2 - AC$ is positive or negative; hence one or else all of the quantities e_1, e_2, e_3 will be necessarily negative, and the intersections will be all real or all imaginary, according as all three are negative or only one is so. A cubic equation in e may be formed containing e_1, e_2, e_3 as its roots by eliminating between the equations

$$e = AC - B'^2, \qquad \square(\lambda U + \mu V) = 0,$$

and the conditions for the reality of the intersections will be that all four coefficients of this cubic shall be of the same sign, which in reality amount only to two, since the first and last must in all cases have the same sign. p. 122

The same objection however of want of symmetry and consequent irrelevancy and complexity attaches to this as much as to the method originally proposed. The following treatment of the question relieves the objection of want of symmetry as far as the coefficients of the same equation are concerned, but in its practical application necessitates an arbitrary and therefore unsymmetrical election to be made between the two sets of coefficients appertaining to the two equations. It is however, I think, too curious and suggestive to be suppressed.

I observe that if the four intersections are all real, an imaginary conic cannot be drawn through them; for the equation to an imaginary conic may always be reduced to the form $Ax^2 + By^2 + Cz^2 = 0$, where A, B, C are all positive and can therefore have at utmost one real point. Consequently the case of total non-intersection is distinguishable from that of complete intersection by the peculiarity that in the one case μ may be so taken that $U + \mu V = 0$ shall represent an imaginary conic, that is, $U + \mu V$ will be a function whose sign never changes for real values of ξ, η, ζ, whereas in the latter case no value of μ will make $U + \mu V = 0$ the equation to an imaginary conic, and therefore $U + \mu V$ will have values on both sides of zero. On the other hand, it is obvious that an infinite number of real as well as unreal conics may be drawn through four imaginary points of intersection. Consequently if we make $U + \mu V = 0$ (supposing the intersections of U and V to be imaginary), there will be a range or ranges of values of μ consistent, and another range or ranges of values of μ inconsistent with real values of ξ, η, ζ; in other words, $U + \mu V = 0$ treated as an equation between the four variables ξ, η, ζ, μ, will give one or more maxima or minima values of μ, in the case supposed, but no such values when the intersections are two or all of them real.

To determine these values of μ, let $d\mu = 0$; then we have

$$\frac{d}{d\xi}(U - \mu V) = 0, \qquad \frac{d}{d\eta}(U - \mu V) = 0, \qquad \frac{d}{d\zeta}(U - \mu V) = 0,$$

that is

$$\Box_{\xi\eta\zeta}(U - \mu V) = 0.$$

In order that any value of μ found from this equation may be a maximum or minimum, Lagrange's condition requires that

$$\left(h\frac{d}{d\xi} + k\frac{d}{d\eta} + l\frac{d}{d\zeta}\right)^2 \mu$$

may be a function of unchangeable sign. p. 123

Now

$$\frac{dU}{d\xi} = \mu\frac{dV}{d\xi} + V\frac{d\mu}{d\xi};$$

therefore since $d\mu = 0$,

$$\frac{d^2U}{d\xi^2} = \mu\frac{d^2V}{d\xi^2} + V\frac{d^2\mu}{d\xi^2}.$$

Hence

$$\frac{d^2\mu}{d\xi^2} = \frac{1}{V}\left(\frac{d}{d\xi}\right)^2 \{U - \mu V\};$$

similarly

$$\frac{d}{d\xi}\frac{d}{d\eta} = \frac{1}{V}\frac{d}{d\xi}\frac{d}{d\eta}\{U - \mu V\},$$

and so on. Making now as before

$$U = a\xi^2 + b\eta^2 + \&c., \qquad V = \alpha\xi^2 + \beta\eta^2 + \&c.,$$

$$a - \mu\alpha = A, \qquad b - \mu\beta = B, \qquad \&c.,$$

the condition for μ, a root of $\square\{U - \mu V\} = 0$, giving μ a maximum or minimum, may be expressed by saying that

$$Ah^2 + Bk^2 + Cl^2 + 2A'kl + 2B'hl + 2C'hk$$

shall be unchangeable in sign for all real values of h, k, l.

The above quantity, by virtue of the equation $\square = 0$, is always the product of two linear functions. Hence we see, as above indicated, that if all these pairs are real, that is, if all the points of intersection of U and V are real, there is no maximum or minimum value of μ; but if only one pair be real and the other two pairs be imaginary, that is, if all the four intersections are imaginary, then two of the values of μ, namely those corresponding to the imaginary pairs, are real maxima or minima values of μ, but the third is illusory.

Now I shall show that if $V = 0$ is a real conic, but the intersections of U and V are all unreal, the value of μ which makes $U + \mu V$ the product of real linear functions of ξ, η, ζ, is always one or the other extreme of the three values of μ which satisfy the equation

$$\square(U - \mu V) = 0.$$

Assume as the three axes of coordinates the three lines joining the vertices of the quadrangle each with each, the two non-intersecting conics may evidently be written under the form

$$U = c(x^2 + y^2) - e(y^2 + z^2) = 0, \qquad V = -\gamma(x^2 + y^2) + \epsilon(y^2 + z^2) = 0;$$ p. 124

these equations being only other modes of writing

$$U = Ax^2 + By^2 + Cz^2, \qquad V = A'x^2 + B'y^2 + C'z^2,$$

in which $A, B, C; A', B', C'$ will be real, because by hypothesis $\square(U + \mu V) = 0$ has all its roots real.

Hence x, y, z are linear functions of ξ, η, ζ and consequently, by a simple inference from a theorem of Prof. Boole[1], the roots of $\square_{\xi\eta\zeta}\{U + \mu V\}$ are identical with those of

$$\square_{xyz}\{U + \mu V\} = 0.$$

These latter are evidently

$$\frac{c}{\gamma}, \qquad \frac{e}{\epsilon}, \qquad \frac{c - e}{\gamma - \epsilon},$$

the third of which is the one which makes $U + \mu V$ the product of two real linears, for we have

$$\gamma U + cV = (c\epsilon - \gamma e)(y^2 + z^2),$$

[1]See Postscript.

$$\epsilon U + eV = (\epsilon c - e\gamma)(x^2 + y^2),$$

$$(\gamma - \epsilon)U + (c - e)V = (c\epsilon - e\gamma)(z^2 - x^2)^2.$$

Now

$$\frac{c}{\gamma} - \frac{c-e}{\gamma-\epsilon} = \frac{e\gamma - c\epsilon}{\gamma(\gamma-\epsilon)}, \qquad \frac{e}{\epsilon} - \frac{c-e}{\gamma-\epsilon} = \frac{e\gamma - c\epsilon}{\epsilon(\gamma-\epsilon)};$$

and ϵ, γ are supposed to have the same sign, as otherwise V would be an unreal conic; hence the ascending or descending order of magnitudes of the three values of λ follows the scale

$$\frac{c}{\gamma}, \quad \frac{e}{\epsilon}, \quad \frac{c-e}{\gamma-\epsilon},$$

as was to be shown.

Imagine now lengths reckoned on a line corresponding to all values of μ from $-\infty$ to $+\infty$, and mark off upon this line by the letters A, B, C, the lengths corresponding with the three roots of $\Box(U + \mu V) = 0$. Then observing that when $\mu = \pm\infty$, $U + \mu V$ is of the same nature as V, and is therefore a possible conic by hypothesis, and agreeing to understand by a possible and impossible region of μ, a range of values for which $U + \mu V$ corresponds to a possible and impossible conic respectively, one or the other of the annexed schemes will represent the circumstances of the case supposed:

$-\infty$ Poss. Reg. A Imposs. Reg. B Poss. Reg. C Poss. Reg. $+\infty$,

$-\infty$ Poss. Reg. A Poss. Reg. B Imposs. Reg. C Poss. Reg. $+\infty$.

But in either scheme it is essential to observe that the *middle* root of $\Box(U+\mu V) = 0$ divides a possible from an impossible region; and therefore p. 125

if we can find u, v, any two values lying between the first and second and second and third roots of the above equation arranged in order of their magnitude, one of the two equations $U + vV = 0$, $U + uV = 0$, will represent a possible and the other an impossible conic: one such couple of values may always be found by taking the roots of the quadratic equation

$$\frac{d}{d\mu}\Box\{U + \mu V\} = 0.$$

Hence calling the two roots thereof m and M, we see (which is in itself a theorem) that one at least of the conics $U + mV = 0$, $U + MV = 0$, must be a possible conic, provided only that $V = 0$ be a possible conic: if both $U + mV$ and $U + MV$ are possible conics, the intersections of U and V are all real, and if not, not[3].

[2] $z^2 - x^2 = 0$ of course represents a real pair of lines.

[3] It must be well observed however that the possibility of the conics $U + mV$ and of $U + MV$ does not imply the reality of the intersections unless the conic V is known to be possible. For if V be impossible ϵ and γ have opposite signs, and therefore $\frac{c-e}{\gamma-\epsilon}$ is intermediate between $\frac{c}{\epsilon}$ and

The criteria for distinguishing possible from impossible conics being well known need not be stated in this place.

We may of course proceed analogously by forming the two conics $lU + V$, $LU + V$, where l and L are roots of

$$\frac{d}{d\lambda}\Box\{\lambda U + V\} = 0$$

upon the supposition of $U = 0$ being a possible conic.

If either of the two U and V be not possible, their intersections are of course impossible, and the question is already decided.

It will be seen as pre-indicated that this method only fails in symmetry because of the choice between the couples m, M, and l, L. But moreover a perfect method for the discrimination of the two cases of unmixed intersection one from the other should (perhaps?) require the application of only a single test (in lieu of the two conditions which the above method supposes), over and above the condition which expresses the fact of the intersections being so unmixed. Such more perfect method I have not yet been able to achieve.

Another interesting question of intersections remains to be discussed, namely, supposing the two conics are known to be non-intersecting, how are we to ascertain if they are external to one another, or if one contains the other? In order to settle this point we must first establish a criterion for determining whether a given *point* is internal or external to a given conic; the point being in general said to be external when two real tangents can be drawn from it to the curve, and internal when this cannot be done. p. 126

Let now

$$\phi(x, y, z) = ax^2 + by^2 + cz^2 + 2a'yz + 2b'zx + 2c'xy = 0,$$

be the equation to any conic: l, m, n the coordinates of any point. Let

$$\begin{aligned} A &= bc - a'^2, \quad & B &= ca - b'^2, \quad & C &= ab - c'^2, \\ A' &= aa' - b'c', \quad & B' &= bb' - c'a', \quad & C' &= cc' - a'b'. \end{aligned}$$

Then the reciprocal equation to the conic is

$$A\xi^2 + B\eta^2 + C\zeta^2 + 2A'\eta\zeta + 2B'\zeta\xi + 2C'\xi\eta = 0,$$

and in making $l\xi + m\eta + n\zeta = 0$, the ratios of ξ, η, ζ must be real if the tangents

$\frac{\gamma}{\epsilon}$, and the scheme for μ will be as here annexed:

$-\infty$ Impossible. A Possible. B Possible. C Impossible. $+\infty$,

so that $U + mV$ and $U + MV$ will both represent possible conics.

drawn from l, m, n are real: this will be found to imply that the determinant

$$\begin{vmatrix} A & C' & B' & l \\ C' & B & A' & m \\ B' & A' & C & n \\ l & m & n & 0 \end{vmatrix}$$

shall be negative[4]. This determinant may be shown[5] to be equal to the product of the determinant

$$\begin{vmatrix} a & c' & b' \\ c' & b & a' \\ b' & a' & c \end{vmatrix}$$

by the quantity

$$al^2 + bm^2 + cn^2 - 2a'mn - 2b'ln - 2c'lm,$$

that is, equal to $\phi(l, m, n) \times \square$.

Hence l, m, n is internal or external to $\phi(x, y, z)$ according as $\phi(l, m, n)$ and $\square\phi$ have the same or contrary sign.

If $\phi(l, m, n) = 0$, the point lies on the conic, and the point is neither internal nor external; if $\square\phi = 0$, the conic becomes a pair of straight lines, and no point can be said either to be within or without such a system. Hence our criterion fails, as it ought to do, just in the very two cases where the distinction vanishes. I believe that this criterion is here given for the first time. p. 127

To return to the two non-intersecting conics. Let us again throw them under the form

$$U = (x^2 + y^2) - e^2(z^2 + y^2), \qquad V = k(x^2 + y^2) - k\epsilon^2(z^2 + y^2),$$

e and ϵ being real, that is, U and V being both functions corresponding to possible conics. Suppose U external to V; then *any point* in U is an external point to V.

Take in U either of the two points represented by the equations $y = 0$, $x^2 = e^2z^2$; substituting these values of y and x, V becomes $k(e^2 - \epsilon^2)z^2$, and $\square V$ becomes $-k^3\epsilon^2(1 - \epsilon^2)$; therefore $(1 - \epsilon^2)(e^2 - \epsilon^2)$ must be positive, that is, ϵ^2 must be one of the extremes of the three values $1, e^2, \epsilon^2$. In like manner, if V

[4]See theorem of the "Diminished Determinant" in Postscript to this paper.

[5]As we know *à priori* by virtue of a theorem given by M. Cauchy, and which is included as a particular case in a theorem of my own, relating to Compound Determinants, that is, Determinants of Determinants, which will take its place as an immediate consequence of my fundamental Theorem given in a Memoir about to appear. The well-known rule for the multiplication of Determinants is also a direct and simple consequence from my theorem on Compound Determinants, which indeed comprises, I believe, in one glance, all the heretofore existing Doctrine of Determinants.

is external to U, e will be also one of the extremes of the same three quantities; and hence, if the two conics are mutually external, unity will be the middle magnitude of the group $e^2, 1, \epsilon^2$.

Now the three roots of $\square(V + \lambda U) = 0$, are

$$\lambda = -k, \qquad \lambda = -k\frac{\epsilon^2}{e^2}, \qquad \lambda = -k\frac{1-\epsilon^2}{1-e^2}.$$

Hence if U and V be without one another, or, as it may be termed, are extra-spatial, the third value of λ will be of a different sign from the first two; but if the two conics be co-spatial, that is, if one includes the other, all the three values of λ will have the same sign. Hence we have the following elegant criterion of co-spatiality of two possible conics expressed by the equations $U = 0$, $V = 0$, between indeterminate coordinates ξ, η, ζ; the coefficients of the cubic function $\square_{\xi\eta\zeta}(\lambda U + \mu V)$ must give only changes or only continuations of sign.

If this test be not satisfied, it will remain to determine which of the two conics contains, and which is contained by the other. Let U contain V, then the order of magnitudes will be $1, e^2, \epsilon^2$; therefore $k\frac{1-\epsilon^2}{1-e^2}$ is greater than k, and therefore $k\frac{1-\epsilon^2}{1-e^2}$, which is that root of the equation $\square(V + \lambda U) = 0$ which is always one or the other of the extremes, is the *greatest* of the three. Hence the scheme for the impossible and possible regions of λ will be as below:

$$-\infty \quad \text{Poss.} \quad A \quad \text{Imposs.} \quad B \quad \text{Poss.} \quad C \quad \text{Poss.} \quad +\infty.$$

Hence if the two roots of $\frac{d}{d\lambda}\{V + \lambda U\} = 0$ be l and L, and of the two conics $V + lU = 0$, $V + LU = 0$, the former be the possible, and the latter the impossible one, U contains V or is contained in it according as l is greater or less than L. p. 128

Observe that if U and V be non-cospatial, so that the three values of μ in $\square(U + \mu V) = 0$ have not all the same sign and consequently zero lies between the greatest and least of them, it will not be necessary to make trial of the characters of the two curves $U + mV = 0$, and $U + MV = 0$, in order to ascertain whether U and V intersect or not; for it will be sufficient to find which of the two quantities m and M substituted for μ in $\square(U + \mu V)$ causes it to have the opposite sign to $\square(U + 0V)$, that is, $\square U$, and this one of the two it is, if either, which will make $U + \mu V$ an impossible conic, and will thus alone serve to determine whether the intersections of U and V are unreal, or the contrary.

It might be a curious question to consider whether, in a certain sense, conics not both possible may not be said to lie one within or without the other. Upon general logical grounds, I think it not improbable that two impossible conics might be discovered each to contain the other; but this is an inquiry which I have not had leisure to enter upon.

I have thus far supposed the roots of $\square(\lambda U + V) = 0$ to be all distinct from one another. I now approach the discussion of the contact of two conics, in which event two or more of the roots will be equal. The condition for simple contact is evidently

$$\Box_{\lambda\mu}\Box_{\xi\eta\zeta}(\lambda U + \mu V) = 0.$$

The unpaired value of λ in $\Box(\lambda U + V)$ makes $\lambda U + V$ an impossible pair of lines, and therefore, in the scheme for λ drawn as above, will separate the possible from the impossible region.

Whether the conics intersect in two real or two unreal points, besides the point of contact, will be known at once by ascertaining whether $U + \mu V = 0$ represents two real or two imaginary lines. If the latter, the two curves lie *dos-à-dos* or one within the other, according as the successions of sign in $\Box(\lambda U + V)$ are all of the same kind or not; if they be all of the same kind, one will include the other, namely, U will include V if the equal roots are greater, and be included in it if they be less than the unequal one. This last conclusion however, it should be observed, is inferred upon the principle of continuity, by making two values of λ approach indefinitely near to one another, but cannot be strictly deduced from the equations given for U and V applicable to the general case, in which the axes of coordinates are the three axes joining the vertices; since these latter, in the case supposed, reduce to two only, and consequently such representation of U and V becomes illusory.

If all three values of λ are equal, the three vertices come together, and hence the two conics will have three consecutive points in common, that is, will have the same circle of curvature. On this supposition the two curves cut at the point of contact, and all four points of intersection are of course real. p. 129

The classification of contacts between two conics may be stated as follows:

Simple contact = one case.

Second degree contact = two cases, namely, common curvature or double contact.

Third degree contact = one case, namely, contact in four consecutive points.

These four cases of course correspond to the several suppositions of there being two equal roots, three equal roots, two pairs of equal roots, or four equal roots in the biquadratic equation obtained between two variables by elimination performed in any manner between the given equations in the two conics.

The first species and the first case of the second species have been already disposed of. I proceed to assign the conditions appertaining to the second case of the second species, when U and V have a double contact.

Let A, A', B, B' be the two pairs of coincident points in which the conics are supposed to meet; either pair of lines $AB, A'B'$, and $AB', A'B$, becomes a coincident pair. Hence such a value of μ can be found as will make $U + \mu V$ the *square of a linear function of* ξ, η, ζ. If therefore we make $U + \mu V = W$, and form the determinant

$$\begin{vmatrix} \dfrac{d^2W}{d\xi^2} & \dfrac{d^2W}{d\xi d\eta} & \dfrac{d^2W}{d\xi d\zeta} & p \\ \dfrac{d^2W}{d\eta d\xi} & \dfrac{d^2W}{d\eta^2} & \dfrac{d^2W}{d\eta d\zeta} & q \\ \dfrac{d^2W}{d\zeta d\xi} & \dfrac{d^2W}{d\zeta d\eta} & \dfrac{d^2W}{d\zeta^2} & r \\ p & q & r & 0 \end{vmatrix}$$

$$= Ap^2 + Bq^2 + Cr^2 + 2Fqr + 2Grp + 2Hpq,$$

where all the coefficients are quadratic functions of μ, and make

$$A = 0, \quad B = 0, \quad C = 0, \quad F = 0, \quad G = 0, \quad H = 0,$$

each of these six equations in μ will have one and the same root in common.

It is, however, enough to select any three; if these vanish together for any value of μ, the remaining three must also vanish. This is a simple application of a general law[6] which will appear in a forthcoming memoir on "Determinants and Quadratic Forms," of which this paper is to be considered as an accidental episode. p. 130

Take now any three of the six equations which for the sake of generality call $P = 0$, $Q = 0$, $R = 0$. The hypothesis of double contact requires that P and Q, Q and R, R and P shall have a factor in common; but these conditions are not sufficiently explicit for our present object, since P, Q, R might be of the form

$$\kappa(\lambda - a)(\lambda - b), \qquad \kappa'(\lambda - b)(\lambda - c), \qquad \kappa''(\lambda - c)(\lambda - a),$$

and would thus satisfy the conditions above stated, without P, Q, R having a common factor. A sufficient criterion is that $fQ + gR$ and P shall have a common factor for all values of f and g.

Let then the resultant of $fQ + gR$ and P be

$$Lf^2 + Mfg + Ng^2;$$

we must have

$$L = 0, \qquad M = 0, \qquad N = 0,$$

where L is the resultant of P and Q, N is the resultant of R and Q, and M is a new function, which if we call $Q = \phi(\lambda)$, $R = \psi(\lambda)$, and suppose a and b to be the two roots of $P = 0$, is easily seen to be equal to

$$\phi a \cdot \psi b + \phi b \cdot \psi a.$$

[6]For statement of this law called the Homaloidal Law, see *Philosophical Magazine* of this month "On Certain Additions, &c." [p. 150 below. Ed.]

This I call the connective of $P \cdot Q$ and $P \cdot R$.

L, M, N may conveniently be denoted by the forms

$$P \cdot Q, \qquad P \cdot R, \qquad Q \cdot P \cdot R.$$

We may now take more generally

$$aP + bQ + cR, \qquad \alpha P + \beta Q + \gamma R,$$

which will have a factor in common for all values of $a, b, c, \alpha, \beta, \gamma$.

I am indebted to Mr Cayley for the remark that the resultant of these two functions is a new quadratic function, which, according to my notation just given, may be put under the form

$$\begin{aligned}&PQ(a\beta - b\alpha)^2 + QR(b\gamma - c\beta)^2 + RP(c\alpha - a\gamma)^2\\ &\quad + PRQ(b\gamma - c\beta)(c\alpha - a\gamma) + QPR(c\alpha - a\gamma)(a\beta - b\alpha)\\ &\quad + RQP(a\beta - b\alpha)(b\gamma - c\beta).\end{aligned}$$

Ternary systems of the six coefficients formed upon the type of (PQ, QR, RP), I call *complete* systems, because the three functions included in such a system equated severally to zero, imply that the remaining three coefficients are all zero. Such a system as (PQ, QR, RP) I term an *incomplete* ternary system as not drawing with it the like implication. Probably (?) we should find on investigation that PRQ, QPR, RQP, would also be an p. 131

incomplete system, but that systems formed after the type of PRQ, RQ, RQP are complete. This however is only matter of conjecture, as I have been too much occupied with other things to enter upon the inquiry. The distinct types of ternary systems are altogether six in number, namely, four of a symmetrical species,

$$\begin{gathered}PQ, \quad QR, \quad RP,\\ PRQ, \quad QPR, \quad RQP,\\ PQ, \quad PQR, \quad QR,\\ PRQ, \quad RQ, \quad RQP;\end{gathered}$$

and two of an unsymmetrical species, namely,

$$\begin{gathered}PQ, \quad PQR, \quad PR,\\ PRQ, \quad RQ, \quad QPR.\end{gathered}$$ [7]

If instead of confining ourselves to three out of the six original quantities, $A, B, C; F, G, H$, we take them all into account, and write down the resultant of

$$aA + bB + cC + fF + gG + hH,$$

[7] PQ, QR, RP, may be compared in a general way with the angles, and PRQ, QPR, RQP, with the sides of a triangle.

$$\alpha A + \beta B + \gamma C + \phi F + \chi G + \eta H;$$

we shall obtain a quadratic function of 15 variables (not however all independent) having 120 coefficients, all of which must be zero. It would be extremely interesting to determine how many *complete* ternary groups can be formed out of these 120 terms.

It will be recollected that we have assigned as the condition of contact in three consecutive points, that a certain cubic equation shall have all its roots real. Now, as well remarked by Mr Cayley, we cannot express this fact by less than three equations in integral terms of the coefficients. Thus if the cubic be written

$$a\lambda^3 + 3b\lambda^2 + 3c\lambda + d = 0,$$

we have as one of such ternary systems,

$$U = ac - b^2 = 0, \qquad V = bd - c^2 = 0, \qquad W = bc - ad = 0.$$

The significant parts of these equations are of course, however, capable of being connected by integral multipliers U', V', W', such that

$$U'U + V'V + W'W = 0.$$

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Any number of functions U, V, W so related, I call *syzygetic* functions, and U', V', W' I term the *syzygetic multipliers*[8]. These in the case supposed are c, a, b, respectively.

In like manner it is evident that the members of any group of functions, more than two in number, whose nullity is implied in the relation of double contact, whether such group form a complete system or not, must be in syzygy.

Thus PQ, PQR, QR, must form a syzygy; nor is there any difficulty in assigning a system of multipliers to exhibit such syzygy. Calling $P = \phi(\lambda)$, $R = \psi(\lambda)$, a and b the two roots of $Q = 0$, I have found that

$$\{(\psi a)^2 + (\psi b)^2\}PQ - (\phi a \cdot \psi a + \phi b \cdot \psi b)PQR + \{(\phi a)^2 + (\phi b)^2\}QR = 0.$$

Again, if we take the *incomplete* system

$$(PQ), \qquad (QR), \qquad (RP),$$

it will be found that

$$L(QR) + M(RP) + N(PQ) = 0,$$

[8]There will be in general various such systems of multipliers.

provided that, calling $a, b; c, d; e, f$, the roots of $P = 0$, $Q = 0$, $R = 0$, respectively, we make

$$\begin{aligned}
L &= (k_0 + k_1 a + k_2 a^2 + k_3 a^3 + k_4 a^4)\frac{(a-c)(a-d)(a-e)(a-f)}{a-b} \\
&\quad + (k_0 + k_1 b + k_2 b^2 + k_3 b^3 + k_4 b^4)\frac{(b-c)(b-d)(b-e)(b-f)}{b-a}, \\
M &= (k_0 + k_1 c + k_2 c^2 + k_3 c^3 + k_4 c^4)\frac{(c-a)(c-b)(c-d)(c-e)(c-f)}{c-d} \\
&\quad + (k_0 + k_1 d + k_2 d^2 + k_3 d^3 + k_4 d^4)\frac{(d-a)(d-b)(d-c)(d-e)(d-f)}{d-c}, \\
N &= (k_0 + k_1 e + k_2 e^2 + k_3 e^3 + k_4 e^4)\frac{(e-a)(e-b)(e-c)(e-d)}{e-f} \\
&\quad + (k_0 + k_1 f + k_2 f^2 + k_3 f^3 + k_4 f^4)\frac{(f-a)(f-b)(f-c)(f-d)}{f-e};
\end{aligned}$$

k_0, k_1, k_2, k_3, k_4 being quite arbitrary, and L, M, N, although presented in a fractional form, being essentially integral.

This fact of L, M, N constituting a system of multipliers to the syzygy QR, RP, PQ, is easily demonstrated; for

$$QR = (c-e)(c-f)(d-e)(d-f),$$

$$RP = (e-a)(e-b)(f-a)(f-b),$$

$$PQ = (a-c)(a-d)(b-c)(b-d).$$

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Hence

$$\begin{aligned}
&L(QR) + M(RP) + N(PQ) \\
&= (a-c)(a-d)(a-e)(a-f)(b-c)(b-d)(b-e)(b-f)(c-e)(c-f)(d-e)(d-f) \\
&\quad \times \sum \frac{k_0 + k_1 a + k_2 a^2 + k_3 a^3 + k_4 a^4}{(a-b)(a-c)(a-d)(a-e)(a-f)} = 0.
\end{aligned}$$

My theory of elimination enables me to explain exactly the nature of L, M, N, and the *reason* of their appearance as syzygetic factors.

Let L_r, M_r, N_r signify what L, M, N become, when all the k's except k_r are taken zero. Then the theory given by me in the *Philosophical Magazine* for the year 1838, or thereabouts[9], shows that $L_0\lambda + L_1$ is the *prime derivee* of the first

[9]This cannot be obtained directly from what is stated in the paper referred to, although contained in the general theory of derivation there given. The arbitrary functions which enter into the expression for the general derivees have been in that paper evaluated only for the prime derivees, which however are only particular phenomena, with reference to the general results of Dialytic Elimination. Hereafter I may give a more general exposition of this remarkable, although ignored or neglected theory. The prime derivees of fx and $f'x$ are Sturm's Functions, cleared of quadratic factors, and are expressed by virtue of the general theorems there laid down as functions of x and of symmetrical functions of the roots of fx. [p. 40 above. Ed.]

degree between the two equations P and $Q \times R$, or, in other words, will be the remainder integralized of

$$\frac{QR}{P}.$$

In like manner $M_0\lambda + M_1$, $N_0\lambda + N_1$ are the integralized remainders of

$$\frac{RP}{Q} \quad \text{and of} \quad \frac{PQ}{R}$$

respectively.

If now the resultant of P, Q and of Q, R are each zero, but the resultant of P and R is not zero, it will be evident that P, Q, R must be of the form

$$f(\lambda + a)(\lambda + c), \qquad g(\lambda + c)(\lambda + d), \qquad h(\lambda + d)(\lambda + b);$$

and therefore $P \times R$ will contain Q, and consequently we must have

$$M_0 = 0, \qquad M_1 = 0.$$

More generally, if we write

$$Q = 0, \qquad \lambda Q = 0, \qquad \lambda^2 Q = 0, \qquad P \times R = 0,$$

and eliminate dialytically, that is, treating $\lambda^4, \lambda^3, \lambda^2, \lambda$ as distinct quantities, we shall obtain

$$\lambda^4 : \lambda^3 : \lambda^2 : \lambda : 1 :: M_4 : M_3 : M_2 : M_1 : M_0;$$

and therefore when $P \times R$ contains Q,

$$M_0 = 0, \quad M_1 = 0, \quad M_2 = 0, \quad M_3 = 0, \quad M_4 = 0.$$

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In like manner, when $Q \times P$ contains R,

$$N_0 = 0, \quad N_1 = 0, \quad N_2 = 0, \quad N_3 = 0, \quad N_4 = 0;$$

and when $R \times Q$ contains P,

$$L_0 = 0, \quad L_1 = 0, \quad L_2 = 0, \quad L_3 = 0, \quad L_4 = 0.$$

Accordingly, we see from the equation

$$L(QR) + M(RP) + N(PQ) = 0,$$

that if $QR = 0$, $RP = 0$; but PQ not $= 0$, then $N = 0$; and therefore

$$N_0 = 0, \quad N_1 = 0, \quad N_2 = 0, \quad N_3 = 0, \quad N_4 = 0,$$

and so in like manner for the remaining corresponding two suppositions[10].

Before proceeding to consider the remaining case of the highest species of contact, I must observe that besides the equations involved in the condition that $A, B, C; F, G, H$, or, which is the same thing, that any three of them shall all have a factor in common, we must have $\square(U + \lambda V)$ containing the square of such common factor. In the memoir before adverted to a general theorem will be given and proved, which shows that this latter condition is involved in the former one; in fact, more generally (but still only as a particular case) that when U and V are quadratic functions of n letters, but $U + eV$ admits of being represented as a complete function of $(n - 2)$ quantities only, which are themselves linear functions of the n letters, then $\square(U + \lambda V)$, which is of course a function of λ of the nth degree, will contain the factor $(\lambda - e)^2$.

When the two conics have four consecutive points in common, the characters of double-point contact and of contact in three consecutive points must exist simultaneously; and consequently the factor common to $A, B, C; F, G, H$, will enter not as a binary but as a ternary factor into $\square(U + \lambda V)$. This gives the extra condition required. As an example take the two conics,

$$U = \frac{y^2}{1-k} + x^2 - z^2 = 0,$$

$$V = y^2 + x^2 - 2kxz + (2k-1)z^2 = 0,$$

$$U + \lambda V = \left(\frac{1}{1-k} + \lambda\right) y^2 + (1+\lambda)x^2 - \{1 + \lambda(1-2k)\}z^2 - 2k\lambda xz.$$

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The complete determinant of $U + \lambda V$ is then

$$-\frac{1}{1-k}\{1 + (1-k)\lambda\}\{(1+\lambda)^2 - 2k\lambda(1+\lambda) + k^2\lambda^2\} = -\frac{1}{1-k}\{1 + (1-k)\lambda\}^3.$$

A, B, C are the determinants of $U + \lambda V$, when $x = 0, y = 0, z = 0$, respectively. Thus

$$A = \left(\frac{1}{1-k} + \lambda\right)(1+\lambda),$$

$$B = \left(\frac{1}{1-k} + \lambda\right)\{1 + \lambda(1-2k)\},$$

[10] Since we are able to assign the values of the syzygetic multipliers in the equations

$$\begin{gathered} L(PQ) + M(QR) + N(RP) = 0, \\ L'(PQ) + M'(PQR) + N'(QR) = 0, \\ L''(QR) + M''(QRP) + N''(RP) = 0, \\ L'''(RP) + M'''(RPQ) + N'''(PQ) = 0, \end{gathered}$$

it follows that we may eliminate between these four equations any three of the six quantities $(PQ), (PRQ)$, &c., and thus express any one of them in terms of any two others: this method, however, is not practically convenient. I may probably hereafter return to this subject.

$$C = k^2\lambda^2 - (1+\lambda)\{1+\lambda(1-2k)\} = \lambda^2(1-k)^2 - 2\lambda(1-k) - 1;$$

$$\lambda = -\frac{1}{1-k}$$

makes $A = 0$, $B = 0$, $C = 0$, and the factor $\lambda + \frac{1}{1-k}$ enters cubed into $\Box(U + \lambda V)$.

Hence the two conics have a contact of the third order.

This is easily verified; for if we pass from general to Cartesian and rectangular coordinates, and make z unity; $U = 0$ will represent an ellipse with centre at the origin, eccentricity $\sqrt{k}$, and mean focal distance 1, and $V = 0$ the circle of curvature at the extremity of the axis major[11].

I had intended to have added some other remarks connected with the present discussion, and also to have appended an *à posteriori* proof of the propositions relative to the reality and otherwise of the vertices and chordal pairs of intersection which I have, at the commencement of this paper, deduced quite legitimately, but in a manner not at first sight perhaps easily intelligible, from the general principles of conjugate forms; but this discussion has run on already to a length so much greater than I had anticipated and than the importance of the inquiry may seem to justify, that I must reserve for a future number of the *Journal* what further matter I may have to communicate concerning it.

Postscript

As I have alluded to Professor Boole's theorem relative to Linear Transformations, it may be proper to mention my theorem on the subject, which is of a much more general character, and includes Mr Boole's (so far as it refers to Quadratic Functions) as a corollary to a particular case. The demonstration will be given in the forthcoming memoir above alluded to.

Let U be a quadratic function of any number of letters $x_1, x_2 \dots x_n$, and let any number r of linear equations of the general form

$$^1a_r x_1 + {}^2a_r x_2 + \cdots + {}^na_r x_n = 0$$

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be instituted between them: and by means of these equations let U be expressed as a function of any $(n-r)$ of the given letters, say of $x_{r+1}, x_{r+2} \dots x_n$, and let U, so expressed, be called M. Let

$$^1a_r x_1 + {}^2a_r x_2 + \cdots + {}^na_r x_n$$

be called L_r. Then the determinant of M in respect to the $(n-r)$ letters above given is equal to the determinant of

$$U + L_1 x_{n+1} + L_2 x_{n+2} + \cdots + L_r x_{n+r},$$

[11] We have thus discussed all the four cases of biconical contact: for an exactly parallel discussion of the theory of contact of a plane with the curve of double curvature in which two surfaces of the second order intersect, see the paper in the *Philosophical Magazine* for this month, before referred to. [p. 148 below. Ed.]

considered as a function of the $(n+r)$ letters

$$x_1 x_2 \dots x_{n+r},$$

divided by the square of the determinant

$$\begin{vmatrix} {}^1a_1 & {}^2a_1 & \cdots & {}^ra_1 \\ {}^1a_2 & {}^2a_2 & \cdots & {}^ra_2 \\ \cdots & \cdots & \cdots & \cdots \\ {}^1a_r & {}^2a_r & \cdots & {}^ra_r \end{vmatrix}.$$

This I call the theorem of Diminished Determinants.

If now we have U a function of r letters, and V of r other letters, and V is derived from U by linear transformations, that is, by r equations connecting the $2r$ letters; then, since U may be considered as a function of *all the* $2r$ letters with abortive coefficients for all the terms where any of the second set of r letters enter, we may apply our theorem of diminished determinants to the question so considered, and the result may be found to represent Mr Boole's theorem in a form rather more general and symmetrical, but substantially identical with that given by Mr Boole.

Thus suppose $\frac{1}{2}ax^2 + bxy + \frac{1}{2}cy^2$ say P, and $\frac{1}{2}\alpha u^2 + \beta uv + \frac{1}{2}\gamma v^2$ say Q, are mutually transformable by virtue of the linear equations

$$lx + my = \lambda u + \mu v, \qquad l'x + m'y = \lambda' u + \mu' v,$$

P may be considered as a function of x, y, u, v, and Q as the value of P, when we eliminate x and y by virtue of the two linear equations

$$L_1 = lx + my - \lambda u - \mu v = 0, \qquad L_2 = l'x + m'y - \lambda' u - \mu' v = 0;$$

we have therefore by our theorem the determinant of Q equal to the squared reciprocal of the determinant

$$\begin{vmatrix} l & m \\ l' & m' \end{vmatrix}$$

multiplied by the determinant

$$\begin{vmatrix} a & b & 0 & 0 & l & l' \\ b & c & 0 & 0 & m & m' \\ 0 & 0 & 0 & 0 & -\lambda & -\lambda' \\ 0 & 0 & 0 & 0 & -\mu & -\mu' \\ l & m & -\lambda & -\mu & 0 & 0 \\ l' & m' & -\lambda' & -\mu' & 0 & 0 \end{vmatrix}.$$

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which last determinant is evidently equal to the determinant of P multiplied by the square of the determinant

$$\begin{vmatrix} \lambda & \mu \\ \lambda' & \mu' \end{vmatrix}.$$

Whence we see that the determinant of Q divided by the square of

$$\begin{vmatrix} \lambda & \mu \\ \lambda' & \mu' \end{vmatrix}$$

is equal to the determinant of P divided by the square of

$$\begin{vmatrix} l & m \\ l' & m' \end{vmatrix}.$$

There is also another way more simple, but less direct, by means of which the theorem of diminished determinants may be made to yield Mr Boole's theorem of transformation[12]. Some unavowed use has been made in the foregoing pages of this former theorem, one of the highest importance in the analytical and geometrical theory of quadratic functions. It has been nearly a year in my possession, and I trust and believe that I am committing no act of involuntary misappropriation in announcing it as a result of my own researches.

[12]Namely, by considering P and Q as each derived from some common function of x, y, u, v, w, by means of the equations $L_1 = 0, L_2 = 0$; the law of Diminished Determinants will then indicate the determinants of P and Q, each under the form of fractions having the same numerator, but whose denominators will be $\begin{vmatrix} \lambda & \mu \\ \lambda' & \mu' \end{vmatrix}$ and $\begin{vmatrix} l & m \\ l' & m' \end{vmatrix}$ respectively.

23.

An Instantaneous Demonstration of Pascal's Theorem by the Method of Indeterminate Coordinates

[Philosophical Magazine, XXXVII. (1850), p. 212]

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The new analytical geometry consists essentially of two parts—the one determinate, the other indeterminate.

The determinate analysis comprehends that class of questions in which it is necessary to assume *independent* linear coordinates, or else to take cognizance of the equations by which they are connected if they are not independent. The indeterminate analysis assumes at will any number of coordinates, and leaves the relations which connect them more or less indefinite, and reasons chiefly through the medium of the general properties of algebraic forms, and their correspondencies with the objects of geometrical speculation. Pascal's theorem of the mystic hexagon, and the annexed demonstration of its fundamental property, belong to this branch of the subject, and afford an instructive and striking example of the application of the pure method of indeterminate coordinates.

Let x, y, z, t, u, v be the sides of a hexagon inscribed in the conic U. Let the hexagon be divided by a new line ϕ in any manner into two quadrilaterals, say $xyz\phi$, $tuv\phi$. Then

$$ay\phi + bvx = U = au\phi + \beta tv;$$

therefore

$$(ay - \alpha u)\phi = \beta tv - bvx;$$

therefore $ay - \alpha u$ and ϕ are the diagonals of the quadrilateral $txvz$.

By construction, ϕ is the diagonal joining x, v (that is, the intersection of x and v) with z, t; and thus we see that $ay - \alpha u$ is the line joining t, x with v, z; but this line passes through y, u. Therefore x, t; y, u; z, v lie in one and the same right line. Q.E.D.

24.

On a New Class of Theorems in Elimination Between Quadratic Functions

[Philosophical Magazine, XXXVII. (1850), pp. 213–218]

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In a forthcoming memoir on determinants and quadratic functions, I have demonstrated the following remarkable theorem as a particular case of one much more general, also there given and demonstrated.

Let U and V be respectively quadratic functions of the same $2n$ letters, and let it be supposed possible to institute n such linear equations between these letters as shall make U and V both simultaneously become identically zero[13]. Then the determinant of $\lambda U + \mu V$, which is of course a function of λ and of μ of the $2n$th degree, will become a *perfect square* of a function of λ and μ of the nth degree; and conversely, if this determinant be a perfect square, U and V may be made to vanish simultaneously by the institution of n linear equations between the $2n$ letters[14].

Let now P and Q be respectively quadratic functions of three letters only, say x, y, z; and let

$$U = P + (lx + my + nz)t,$$
$$V = Q + k(lx + my + nz)t.$$

The determinant of $\lambda U + \mu V$ in respect to x, y, z, t is easily seen to be $(\lambda + k\mu)^2$ times the determinant of

$$\lambda P + \mu Q + (lx + my + nz)t$$

in respect to x, y, z, t. Hence if we call

$$\lambda P + \mu Q + (lx + my + nz)t = W,$$

and make $\Box_{xyzt} W$ a squared function of λ, μ or which is the same thing, if

$$\Box_{\lambda\mu}\Box_{xyzt}\{W\} = 0,$$

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U and V will vanish simultaneously when two linear relations are instituted between the quantities (all or some of them) x, y, z, t.

In order that this may be the case, it will be seen to be sufficient that

$$P = 0, \qquad Q = 0, \qquad lx + my + nz = 0,$$

[13] In the more general theorem above alluded to, the number of letters is any number m, the number of linear equations being any number not exceeding $\frac{m}{2}$.

[14] When $n = 1$, we obtain a theorem of elimination between two quadratics, which has been already given by Professor Boole.

shall coexist; for then two equations between x, y, z of which $lx + my + nz = 0$ will be one, will suffice to make U and V each identically zero. Hence we have the following theorem:

$$\Box_{\lambda\mu}\Box_{xyzt}\{\lambda U + \mu V + (lx + my + nz)t\}$$

is a factor of the resultant of

$$P = 0, \qquad Q = 0, \qquad lx + my + nz = 0.$$

A comparison of the orders of the resultant and the determinant shows that they must be identical, à-ci-près, of a numerical factor, which, if the resultant be taken in its *general* lowest terms, may no doubt be easily shown to be unity.

As an illustration of our theorem, let

$$P = xy + yz + zx, \qquad Q = cxy + ayz + bzx.$$

Then

$$\Box_{xyzt}\{\lambda P + \mu Q + (lx + my + nz)t\} = \begin{vmatrix} 0 & \lambda + c\mu & \lambda + b\mu & l \\ \lambda + c\mu & 0 & \lambda + a\mu & m \\ \lambda + b\mu & \lambda + a\mu & 0 & n \\ l & m & n & 0 \end{vmatrix}$$

$$\begin{aligned} &= n^2(\lambda + c\mu)^2 + m^2(\lambda + b\mu)^2 + l^2(\lambda + a\mu)^2 \\ &\quad - 2lm(\lambda + b\mu)(\lambda + a\mu) - 2mn(\lambda + c\mu)(\lambda + b\mu) \\ &\quad - 2nl(\lambda + a\mu)(\lambda + c\mu) \\ &= \lambda^2\{n^2 + m^2 + l^2 - 2lm - 2mn - 2nl\} \\ &\quad + 2\lambda\mu\{cn^2 + bm^2 + al^2 - lm(a + b) - mn(b + c) - nl(c + a)\} \\ &\quad + \mu^2\{c^2n^2 + b^2m^2 + a^2l^2 - 2ablm - 2bcmn - 2canl\}. \end{aligned}$$

And we thus obtain, finally,

$$\begin{aligned} &\Box_{\lambda\mu}\Box_{xyzt}\{\lambda P + \mu Q + (lx + my + nz)t\} \\ &= (n^2 + m^2 + l^2 - 2lm - 2mn - 2nl) \\ &\qquad \times (c^2n^2 + b^2m^2 + a^2l^2 - 2ablm - 2bcmn - 2canl) \\ &\quad - \{cn^2 + bm^2 + al^2 - lm(a + b) - mn(b + c) - nl(c + a)\}^2 \\ &= -4lmn\{(a - b)(a - c)l + (b - a)(b - c)m + (c - a)(c - b)n\}. \end{aligned}$$

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Now to obtain the resultant of

$$xy + yz + zx = 0, \qquad cxy + azy + bxz = 0, \qquad lx + my + nz = 0,$$

we need only find the four systems *in their lowest terms* of $x : y : z$, which satisfy the first two equations, and multiply the four linear functions obtained by substituting these values of x, y, z in the fourth: the product will contain the resultant of the system affected with some numerical factor. In the present case, the four systems of x, y, z are

$$x = 0, \quad y = 0, \quad z = 1,$$

$$y = 0, \quad z = 0, \quad x = 1,$$

$$z = 0, \quad x = 0, \quad y = 1,$$

$$x = (a - b)(a - c), \quad y = (b - a)(b - c), \quad z = (c - a)(c - b),$$

and accordingly the product of

$$\begin{aligned} &lx_1 + my_1 + nz_1, \\ &lx_2 + my_2 + nz_2, \\ &lx_3 + my_3 + nz_3, \\ &lx_4 + my_4 + nz_4, \end{aligned}$$

becomes

$$lmn\{(a - b)(a - c)l + (b - a)(b - c)m + (c - a)(c - b)n\},$$

agreeing with the result obtained by my theorem,—a special numerical factor 4, arising from the peculiar form of the equations, having disappeared from the resultant.

A geometrical demonstration may be given of the theorem which is instructive in itself, and will suggest a remarkable extension of it to functions containing more than three letters; the equation

$$\Box_{xyzt}\{\lambda U + \mu V + (lx + my + nz)t\} = 0,$$

which is a quadratic equation in $\lambda : \mu$, may easily be shown to imply that the conic $\lambda U + \mu V$ is touched by the straight line

$$lx + my + nz = 0.$$

And we thus see that in general two conics,

$$\lambda U + \mu V = 0,$$

passing through the intersections of two given conics,

$$U = 0, \qquad V = 0,$$

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may be drawn to touch a given line. If, however, the given line passes through any of the four points of intersection, in such case only one conic can be drawn to touch it; accordingly

$$\square\square\{\lambda U + \mu V + (lx + my + nz)t\}$$

must be zero when l, m, n are so taken as to satisfy this condition, that is, if

$$lx_1 + my_1 + nz_1 = 0,$$

or

$$lx_2 + my_2 + nz_2 = 0,$$

or

$$lx_3 + my_3 + nz_3 = 0,$$

or

$$lx_4 + my_4 + nz_4 = 0,$$

whence the theorem.

Now suppose U and V to be each functions of four letters, x, y, z, t; when

$$\square_{xyztu}\{\lambda U + \mu V + (lx + my + nz + pt)u\} = 0,$$

the conoid $\lambda U + \mu V$ touches the plane

$$lx + my + nz + pt = 0;$$

and $\square = 0$ being a cubic equation, in general three such conoids can be drawn.

Considerations of analogy make it obvious to the intuition, that in the particular case of two of these becoming coincident, the given plane

$$lx + my + nz + pt$$

must be a tangent plane to those two coincident conoids at one of the points where it meets the intersections of $U = 0, V = 0$; that is

$$lx + my + nz + pt = 0$$

will pass through a tangent line to, or in other words, may be termed a tangent plane to the intersections. Hence the following analytical theorem, derived from supposing q, r, s, t to be proportional to the areas of the triangular faces of the pyramid cut out of space by the four coordinate planes to which x, y, z, t refer. As these planes are left indefinite, q, r, s, t are perfectly arbitrary.

Theorem. The resultant of

$$\left.\begin{array}{ll} 1. & U = 0 \\ 2. & V = 0 \end{array}\right\} \quad \text{where } U \text{ and } V \text{ are functions of } x, y, z, t;$$

and

$$3. \qquad lx + my + nz + pt = 0;$$

together with

$$4. \qquad \begin{vmatrix} \frac{dU}{dx} & \frac{dU}{dy} & \frac{dU}{dz} & \frac{dU}{dt} \\ \frac{dV}{dx} & \frac{dV}{dy} & \frac{dV}{dz} & \frac{dV}{dt} \\ l & m & n & p \\ q & r & s & t \end{vmatrix} = 0;$$

which system, it will be observed, consists of three quadratic functions, and one linear function of x, y, z, t, contains the factor p. 143

$$\square_{\lambda\mu}\square_{xyzt}\{\lambda U + \mu V + (lx + my + nz + pt)u\}.$$

This last quantity is of the 4×3th, that is, the 12th order in respect of the coefficients in U and V combined; of the 4×2th, that is, the 8th order in respect of l, m, n, p; and of the zero order in respect of q, r, s, t.

The resultant which contains it is of the $(4 + 4 + 2 \cdot 4)$th, that is, 16th order in respect to the coefficients in U and V; of the $(4 + 8)$th, that is, the 12th, in respect of l, m, n, p; and of the 4th in respect of q, r, s, t. Hence the special (and, as far as the geometry of the question is concerned, the unnecessary, I may not say extraneous or irrelevant) factor which enters into the resultant is of the 4th order in respect to the combined coefficients of U and V[15]; and of the same order in respect to l, m, n, p, and in respect to q, r, s, t.

I have not yet succeeded in divining its general value.

In the very particular example, of the system,

$$\alpha x^2 + \beta y^2 = 0,$$

$$cz^2 + dt^2 = 0,$$

$$lx + my + nz + pt = 0,$$

$$\begin{vmatrix} \alpha x & \beta y & 0 & 0 \\ 0 & 0 & cz & dt \\ l & m & n & p \\ q & 0 & 0 & 0 \end{vmatrix} = 0,$$

I find that the double determinant is

$$c^2 d^2 \alpha^2 \beta^2 (cp^2 + dn^2)^2 (m^2\alpha + l^2\beta)^2,$$

and the resultant is

$$q^4 c^2 d^2 \alpha^2 \beta^4 (cp^2 + dn^2)^4 (m^2\alpha + l^2\beta)^2,$$

[15]And consequently of the second in respect to the separate coefficients of each.

giving as the special factor

$$q^4\beta^2(cp^2+dn^2)^2.$$

I believe that the theorem which I have here given for determining the condition that $lx+my+nz+pt$ shall be a tangent plane to the intersection of two conoids U and V, namely, that the determinant of

$$\lambda U+\mu V+(lx+my+nz+pt)u$$

shall have two equal roots, is altogether novel. p. 144

What is the meaning of all three roots of this determinant becoming equal, that is, of only one conoid being capable of being drawn through the intersection of U and V to touch the plane

$$lx+my+nz+pt?$$

Evidently (*ex vi analogiae*) that this plane shall pass through three consecutive points of the curve of intersection, that is, that it shall be the osculating plane to the curve.

If we return to the intersection of two co-planar conics, and if we suppose a line to be drawn through two of the points of intersection, the conics capable of being drawn through the four points of intersection to touch the line, besides becoming coincident, evidently degenerate each into a pair of right lines. It would seem, therefore, by analogy, that if a plane be drawn including any two tangent lines to the curve of intersection of two surfaces of the second degree, this should be touched by two coincident cones drawn through the curve of intersection, and consequently every such double tangent plane to the intersection of two conoids (and it is evident that one or more of these can be taken at every point of the curve) must pass through one of the vertices of the four cones in which the intersection may also be considered to lie; and it would appear from this, that in general four double tangent planes admit of being drawn to the curve, which is the intersection of two conoids, at each point thereof. At particular points a tangent plane may be drawn passing through more than one of the vertices, and then of course the number of double tangent planes that can be drawn will be lessened. These results, indicated by analogy, become immediately apparent on considering the curve in question as traced upon any one of the four containing cones. For the plane drawn through a tangent at any point, and the vertex of the cone being a tangent plane to the cone, must evidently touch the curve again where it meets it. We thus have an additional confirmation of the analogy between a point of intersection of two curves and the tangent at any point of the intersection of two surfaces.

I might extend the analytical theorems which have been established for functions of three and four to functions of a greater number of variables; but enough has been done to point out the path to a new and interesting class of theorems at once in elimination and in geometry, which is all that I have at present leisure or the disposition to undertake.

25.

Additions to the Articles "On a New Class of Theorems," and "On Pascal's Theorem"

[Philosophical Magazine, XXXVII. (1850), pp. 363–370]

p. 145

First addition.—I have alluded in the second of the above articles to a more general theorem, comprising, as a particular case, the theorem there given for the simultaneous evanescence of two quadratic functions of $2n$ letters, on n linear equations becoming instituted between the letters.

In order to make this generalization intelligible, I must premise a few words on the Theory of Orders, a term which I have invented with particular reference to quadratic functions, although obviously admitting of a more extended application. A linear function of all the letters entering into a function or system of functions under consideration I call an order of the letters, or simply an order. Now it is clear that we may always consider a function of any number of letters as a function of as many orders as there are letters; but in certain cases a function may be expressed in terms of a fewer number of orders than it has letters, as when the general characteristic function of a conic becomes that of a pair of crossing lines or a pair of coincident lines, in which event it loses respectively one and two orders, and so for the characteristic of a conoid becoming that of a cone, a pair of planes or two coincident planes, in which several events, a function of four letters becomes that of only three orders, or two orders, or one order, respectively. When a function may be expressed by means of r orders less than it contains letters, I call it a function minus r orders. I now proceed to state my theorem.

Let U and V be functions each of the same m letters, and suppose that the determinant in respect of those letters of $U + \mu V$ contains i pairs of p. 146

equal linear factors of μ; then it is possible, by means of i linear equations instituted between the letters, to make U and V each become functions of *the same* $m - 2i$ orders; and conversely, if by i equations between the letters U and V may be made functions of the *same* $m - 2i$ orders, the determinant of $U + \mu V$ considered as a function of μ will contain i square factors.

Thus when $m = 2n$ and $i = n$, U and V will each become functions of zero orders, that is, will both disappear, provided that on the institution of a certain system of n linear equations, among the letters of which U and V are functions, the determinant of $(U + \mu V)$ is a perfect square,—which is the theorem given in the article referred to.

So for example if U and V be quadratic functions of four letters, and therefore the characteristics of two conoids, $\square(U + \mu V)$ being a perfect square, expresses that these conoids have a straight line in common lying upon each of their surfaces.

If U and V be quadratic functions of three letters only, and admit therefore of being considered as the characteristics of two conics, $\square(U + \mu V)$ containing a square factor, is indicative of these conics having a common tangent at a common point, that is, of their touching each other at some point; for it is easily shown that the disappearance of two orders from any quadratic function by virtue of one linear function of its letters being zero, indicates that the line, plane, &c. of which the linear function is the characteristic is a tangent to the curve, surface, &c. of which the quadratic function is the characteristic.

I pass now to a generalization of the theorem which shows how to express, under the form of a double determinant, the resultant of one linear and two quadratic homogeneous functions of three letters (which I should have given in the original paper, had I not there been more intent upon developing an ascending scale than of expatiating upon a superficial ramification of analogies), and which constitutes my *Second addition* to that paper, to wit—

If U and V be homogeneous quadratic, and $L_1, L_2 \ldots L_n$ homogeneous linear functions of $(n + 2)$ letters $x_1, x_2 \ldots x_{n+2}$, the determinant of the entire system of $n + 2$ functions is equal to

$$\square_{\lambda,\mu}\square_{x_1,x_2,\ldots,x_{n+2},t_1,t_2,\ldots,t_n}\{\lambda U + \mu V + L_1 t_1 + L_2 t_2 + \cdots + L_n t_n\};$$

the demonstration is precisely similar to the analytical one given in the September Number[16] for the particular case of $n = 1$.

When $n = 0$, we revert to Mr Boole's theorem of elimination between U and V already adverted to. The proof, it will be easily recognized, does not require the application of the more general theorem relative to the simultaneous p. 147

depression of orders of two quadratic functions, but only the limited one before given, which supplies the conditions of their simultaneous disparition. I now proceed to develop more particularly certain analogies between the theory of the mutual contacts of two conics, and that of the tangencies to the intersection of two conoids.

But here again I must anticipate some of the results which will be given in my forthcoming memoir on Determinants and Quadratic Functions, by explaining what is to be understood by minor determinants, and the relation in which they stand to the complete determinant in which they are included. This preliminary explanation, and the statement of the analogies above alluded to, will constitute my *Third* and last addition.

Imagine any determinant set out under the form of a square array of terms. This square may be considered as divisible into lines and columns. Now conceive any one line and any one column to be struck out, we get in this way a square, one term less in breadth and depth than the original square; and by varying in every possible manner the selection of the line and column excluded, we obtain,

[16] p. 140 above.

supposing the original square to consist of n lines and n columns, n^2 such minor squares, each of which will represent what I term a First Minor Determinant relative to the principal or complete determinant. Now suppose two lines and two columns struck out from the original square, we shall obtain a system of

$$\left\{\frac{n(n-1)}{2}\right\}^2$$

squares, each two terms lower than the principal square, and representing a determinant of one lower order than those above referred to. These constitute what I term a system of Second Minor Determinants; and so in general we can form a system of rth minor determinants by the exclusion of r lines and r columns, and such system *in general* will contain

$$\left\{\frac{n(n-1)\cdots(n-r+1)}{1\cdot 2\cdots r}\right\}^2$$

distinct determinants.

I say "*in general*"; because if the principal determinant be totally or partially symmetrical in respect to either or each of its diagonals, the number of distinct determinants appertaining to each system of minors will undergo a material diminution, which is easily calculable.

Now I have established the following law:—

The whole of a system of rth minors being zero, implies only $(r+1)^2$ equations, that is, by making $(r+1)^2$ of these minors zero, all will become zero; and this is true, no matter what may be the dimensions or form of the complete determinant. But furthermore, if the complete determinant be formed from a quadratic function, so as to be symmetrical about one of its diagonals, then $\frac{1}{2}(r+1)(r+2)$ only of the rth minors being zero, will serve p. 148

to imply that all these minors are zero. Of course, in applying these theorems, care must be taken that the $(r+1)^2$ or $\frac{1}{2}(r+1)(r+2)$ selected equations must be mutually non-implicative, and shall constitute independent conditions.

In the application I am about to make of these principles, we shall have only to deal with a system of *first* minors and of a *symmetrical* determinant. If three of these properly selected be zero, from the foregoing it appears that all must be zero.

Now let U and V be characteristics of two conics, that is, let each be a function of only three letters, it may be shown (see my paper[17] in the *Cambridge and Dublin Mathematical Journal* for November, 1850) that the different species of contacts between these two conics will correspond to peculiar properties of the compound characteristic $U + \mu V$.

If the determinant of this function have two equal roots, the conics simply touch; if it have three equal roots, the conics have a single contact of a higher order,

[17] p. 119 above.

that is, the same curvature; if its six first minors become zero simultaneously for the same value of μ, the conics have a double contact. If the same value of μ, which makes all these first minors zero, be at the same time not merely a double root (as of analytical necessity it always must be) but a treble root of

$$\square(U + \mu V) = 0,$$

then the conics have a single contact of the highest possible order short of absolute coincidence, that is, they meet in four consecutive points.

The parallelism between this theory and that of two quadratic functions P, Q, and one linear function L[18] of four letters, say x, y, z, t, is exact[19]. For let $P + Lu + \mu Q$ be now taken as our compound characteristic (a function, it will be observed, of five letters, x, y, z, t, u); if its determinant have two equal roots, L has two consecutive points in common with the intersection of P and Q, that is, passes through a tangent to that intersection; if it have three equal roots, L has three consecutive points in common with the said intersection, that is, is an osculating plane thereto; if its fifteen first minors admit of all being made simultaneously zero, L has a double contact with the intersection of P and Q, that is, it is a tangent plane to some one of the four cones of the second order containing this intersection; p. 149

if the same linear function of μ which enters into all these first minors be contained cubically in the complete determinant, then the plane L passes through four consecutive points of the intersection of P and Q, and the points where it meets the curve will be points of contrary plane flexure; and, as it seems to me, at such points the tangential direction of the curve must point to the summit of one or other of the four cones above alluded to[20]. In assigning the conditions for L being a double tangent plane to the intersection of P and Q, we may take any three independent minors at pleasure equal to zero. One of these may be selected so as to be clear of the coefficients of L; in fact, the determinant of $P + \mu Q$ will be a first minor of $P + \mu Q + Lu$; μ may thus be determined by a biquadratic equation; and then, by properly selecting the two other minors, we may obtain two equations in which only the first powers of the coefficients of x, y, z, t in L appear, and may consequently obtain L under the form of

$$(ae + \alpha)x + (be + \beta)y + (ce + \gamma)z + (de + \delta)t,$$

[18] Observe that $P = 0, Q = 0, L = 0$ now express the equations to two conoids and a plane respectively.

[19] This parallelism may be easily shown analytically to imply, and be implied, in the geometrical fact, that the contact of the plane L with the intersection of the two surfaces P and Q, is of exactly the same kind as the contact (which must exist) between the two conics which are the intersections of P and Q respectively with the plane L.

[20] If this be so, then we have the following geometrical theorem:—"The summit of one of the four cones of the second degree which contain the intersections of two surfaces of the second order drawn in any manner respectively through two given conics lying in the same plane, and having with one another a contact of the third degree, will always be found in the same right line, namely, in the tangent line to the two given conics at the point of contact."

where $a, \alpha;\ b, \beta;\ c, \gamma;\ d, \delta$ will be known functions of any one of the four values of μ. The point of contact being given will then serve to determine e, and we shall thus have the equation to each of the four double tangent planes at any given point fully determined.

In the foregoing discussions I have freely employed the word *characteristic* without previously defining its meaning, trusting to that being apparent from the mode of its use. It is a term of exceeding value for its significance and brevity. The characteristic of a geometrical figure[21] is the function which, equated to zero, constitutes the equation to such figure. Pluecker, I think, somewhere calls it the line or surface function, as the case may be. Geometry, analytically considered, resolves itself into a system of rules for the construction and interpretation of characteristics. One more remark, and I have done. A very comprehensive theorem has been given at the commencement of this commentary, for interpreting the effect of a complete determinant of a linear function of two quadratic functions $(U + \mu V)$, having p. 150

one or more pairs of equal factors $(e + \epsilon\mu)$. But here a far wider theory presents itself, of which the aim should be to determine the effect and meaning of this determinant, having any amount and distribution of multiplicity whatsoever among its roots. Nor must our investigations end at that point; but we must be able to determine the meaning and effect of common factors, one or more entering into the successive systems of *minor* determinants derived from the complete determinant of $U + \mu V$.

Nor are we necessarily confined to two, but may take several quadratic functions simultaneously into account.

Aspiring to these wide generalizations, the analysis of quadratic functions soars to a pitch from whence it may look proudly down on the feeble and vain attempts of geometry proper to rise to its level or to emulate it in its flights.

The law which I have stated for assigning the number of independent, or to speak more accurately, non-coevanescent determinants belonging to a given system of minors, I call the Homaloidal law, because it is a corollary to a proposition which represents analytically the indefinite extension of a property common to lines and surfaces to all loci (whether in ordinary or transcendental space) of the first order, all of which loci may, by an abstraction derived from the idea of levelness common to straight lines and planes, be called Homaloids. The property in question is, that neither two straight lines nor two planes can have

[21] More generally, the characteristic of any fact or existence is the function which, equated to zero, expresses the condition of the actuality of such fact or existence. Perhaps the most important pervading principle of modern analysis, but which has never hitherto been articulately expressed, is *that*, according to which we infer, that when one fact of whatever kind is implied in another, the characteristic of the first must contain as a factor the characteristic of the second; and that when two facts are mutually involved, their characteristics will be powers of the same integral function. The doctrine of characteristics, applied to dependent systems of facts, admits of a wide development, logical and analytical.

a common segment; in other words, if n independent relations of rectilinearity or of coplanarity, as the case may be, exist between triadic groups of a series of $n+2$, or between tetradic groups of a series of $n+3$ points respectively, then every triad or tetrad of the series, according to the respective suppositions made, will be in rectilinear or in plane order. So, too, if n independent relations of *coincidence* exist between the duads formed out of $n+1$ points, every duad will constitute a coincidence.

This homaloidal law has not been stated in the above commentary in its form of greatest generality. For this purpose we must commence, not with a square, but with an oblong arrangement of terms consisting, suppose, of m lines and n columns. This will not in itself represent a determinant, but is, as it were, a Matrix out of which we may form various systems of determinants by fixing upon a number p, and selecting at will p lines and p columns, the squares corresponding to which may be termed determinants of the pth order. We have, then, the following proposition. The number of uncoevanescent determinants constituting a system of the pth order derived from a given matrix, n terms broad and m terms deep, may equal, but can never exceed the number

$$(n-p+1)(m-p+1).$$

Remark on Pascal's and Brianchon's Theorems

p. 151

I omitted to state, in the September Number of the *Journal*[22], that the demonstration there given by me for Pascal's, applied equally to Brianchon's theorem. This remark is of the more importance, because the fault of the analytical demonstrations hitherto given of these theorems has been, that they make Brianchon's consequence of Pascal's, instead of causing the two to flow simultaneously from the application of the same principles. No demonstration can be held valid in *method*, or as touching the essence of the subject-matter, in which the indifference of the duadic law is departed from. Until these recent times, the analytic method of geometry, as given by Descartes, had been suffered to go on halting as it were on one foot. To Pluecker was reserved the honour of setting it firmly on its two equal supports by supplying the complementary system of coordinates. This invention, however, had become inevitable, after the profound views promulgated by Steiner, in the introduction to his Geometry, had once taken hold of the minds of mathematicians. To make the demonstration in the article referred to apply, *totidem literis*, to Brianchon's theorem (recourse being had to the correlative system of coordinates), it is only needful to consider U as the characteristic of the tangential envelope of the conic, x, y, z, t, u, v as the characteristics of the six points of the circumscribed hexagon, ϕ the characteristic of the point in which the line x, v meets the line z, t; $ay - \alpha u$ will then be shown to characterize the point in which t, x meets v, z; and thus we see that y, u; t, x; v, z,

[22] p. 138 above.

the three pairs of opposite sides of the hexagon, will meet in one and the same point, which is Brianchon's theorem.